

SpecFirst React A11y

Phase 8 Documentation: Verify and Generate Evidence Report

How the pipeline closes: reruns the same tests after Bob's patch, compares to baseline, and produces a human-readable evidence packet

Phase 8 reruns the identical generated tests against the patched component, computes which baseline failures Bob resolved and whether Bob introduced new failures, and writes the final evidence report. The allowed conclusion is always: Passed SpecFirst automated accessibility contract. Manual review required. Never WCAG compliant.

Prepared for the IBM Bob Hackathon workflow

1. Phase 8 in one sentence

Phase 8 re-verifies immutable artifact integrity, reruns the exact same Playwright command used in Phase 6, compares the final test results to the baseline, and generates a human-readable evidence report that respects the claim boundary established in the locked spec.

Pipeline context:

Phase 7: Bob patches component Phase 8: Verify and generate evidence report ← you are here

The central design rule is that Phase 8 uses the same tests as Phase 6. The locked spec, the test file, and the harness must be unchanged. Only the component source is allowed to differ. The comparison between Phase 6 failures and Phase 8 results is what makes Bob's contribution visible and auditable.

2. Purpose

- Confirm the full chain of custody is intact: locked spec, test file, and harness are unchanged since Phase 5.
- Rerun the identical Playwright command to get a post-patch test result.
- Compare final results to the Phase 6 baseline to identify resolved failures, remaining failures, and any new failures Bob introduced.
- Generate evidenceReport.md — a human-readable document that judges and developers can read to understand what the pipeline verified.
- Enforce the claim boundary: the report must never state WCAG compliance, full accessibility, or screen reader compatibility.

3. Inputs

Input artifact	Role in Phase 8
bobPatchResult.json	Must have status patched or bob-session-missing. Provides patch scope result, Bob session path, and confirms integrity passed in Phase 7.
baselineResult.json	Provides the baseline failed check IDs and the Phase 5 hash snapshots for re-verification.
lockedSpec.json	Provides manual review items, accessibility claim boundary, component identity, and the hash expected value.
testGenerationResult.json	Provides test file path, harness file path, and expected hashes.
Generated test file	Re-hashed and re-verified. Then executed by Playwright.
Generated harness file	Re-hashed and re-verified. Required for the Vite demo route.

4. Outputs

Output artifact	Purpose
finalVerificationResult.json	Machine-readable final result. Status, component identity, artifact verification, baseline comparison, final test summary, check comparison, manual review list, claim boundary, and Bob session evidence. This is the definitive pipeline output.
finalOutput.json	Raw Playwright JSON reporter output from the post-patch run.
finalOutput.txt	Raw stderr from the Playwright process, preserved for debugging.
evidenceReport.md	Human-readable evidence packet. The deliverable for judges, reviewers, and developers. Eight sections: component, locked contract, generated tests, baseline result, IBM Bob remediation, final verification, manual review still required, and claim boundary.

5. Scope boundaries

Phase 8 should do:

- Read bobPatchResult.json and refuse to proceed if status does not allow continuation.
- Re-verify SHA-256 hashes of lockedSpec.json, the test file, and the harness file against Phase 5/6 records.
- Rerun the exact same Playwright command as Phase 6 (`npx playwright test <testFile> --reporter=json`).
- Parse the final Playwright output and map results to locked-spec check IDs.
- Compute the resolved and new-failure sets by comparing final failed IDs to baseline failed IDs.
- Write finalVerificationResult.json and evidenceReport.md.
- Include manual review requirements and claim boundary in the report.

Phase 8 should not do:

- Patch the component again or call Bob for additional patching.
- Edit the locked spec, test file, or harness.
- Claim WCAG compliance, full accessibility, or screen reader compatibility.
- Hide remaining failures. If Bob did not fix every check, the report must say so honestly.
- Block on Bob session missing. Evidence completeness is a warning, not a blocker.

6. Artifact integrity re-verification

Phase 8 re-hashes the three immutable artifacts and compares to the Phase 5 baseline hashes stored in baselineResult.json. If any hash changed, Phase 8 produces status invalidated and stops.

Artifact	Expected hash source	On mismatch
lockedSpec.json	baselineResult.lockedSpecHash	status: invalidated, stop
Generated test file	baselineResult.testFileHashBeforeRun	status: invalidated, stop
Generated harness file	baselineResult.harnessFileHashBeforeRun	status: invalidated, stop

The component source file is allowed to have changed. That is what Bob was supposed to do. Only the three artifacts above must be unchanged. If the locked spec changed, the evidence chain is broken regardless of test results.

7. Baseline-to-final comparison

Phase 8 computes three derived sets from the baseline failed check IDs and the final failed check IDs:

Set	Definition	Interpretation
resolved	$\text{baselineFailed} \setminus \text{finalFailed}$	Checks Bob fixed. These passed in Phase 8 but failed in Phase 6.
newFailures	$\text{finalFailed} \setminus \text{baselineFailed}$	Checks Bob broke. These failed in Phase 8 but passed in Phase 6. A non-empty new-failures set is a serious concern.
stillFailing	$\text{baselineFailed} \cap \text{finalFailed}$	Checks Bob did not fix. Reported honestly. Not a blocker for generating the evidence report.

The key hackathon statement from this comparison: *"Bob resolved N baseline failures without introducing new generated-test failures."*

8. Evidence report structure

evidenceReport.md is the human-facing output. It follows a fixed eight-section structure:

Section	Content
1. Component	Name, path, and pattern classification.
2. Locked Contract	Locked spec path, hash, unchanged status, and source model. Confirms the contract was frozen before Bob.
3. Generated Tests	Test file path and hash, harness path and hash, test count. Confirms tests were not edited.
4. Baseline Result	Status red-confirmed. List of checks that failed before Bob. Explains what this means.
5. IBM Bob Remediation	Bob session reference, changed files, patch scope status, artifact integrity table.
6. Final Verification	Final test summary (N passed / N failed), resolved checks, still-failing checks, new failures. The comparison statement.
7. Manual Review Still Required	Full list of manual review IDs from the locked spec. Reminds readers that automated checks do not cover screen reader quality, visual focus, cross-browser AT, cognitive load, or native select preference.
8. Claim Boundary	Allowed conclusion and forbidden conclusions. Taken from the locked spec's accessibilityClaimBoundary.

9. Claim boundary enforcement

The locked spec defines the language boundary. Phase 8 must preserve it in the evidence report without softening or bypassing it.

Category	Examples
Allowed conclusion	Passed SpecFirst automated accessibility contract. Manual review required.
Forbidden claims	WCAG 2.2 AA compliant. Fully accessible. Accessibility guaranteed. Screen reader compatible. Component passed WCAG.

This boundary applies even when all 11 tests pass. Automated testing proves specific contract checks are satisfied by the DOM at the time of the run. It does not prove the component is usable by screen reader users in real-world conditions, across browsers, or under cognitive load. The manual review items in the locked spec name exactly what remains to be verified by humans.

10. Status model

Status	Meaning	nextPhase
passed	All generated tests pass after Bob's patch, immutable artifacts unchanged, patch scope passed. Still includes manual review required.	canContinue true
failed	One or more generated tests still fail after Bob's patch. Evidence report shows remaining failures. Report is still generated.	canContinue true
invalidated	Immutable artifact hashes changed between Phase 5 and Phase 8. Evidence chain is broken. Report is not generated.	canContinue false
infra-failed	Playwright could not run correctly due to server, browser, or import failures.	canContinue false
skipped	bobPatchResult.json status does not allow Phase 8 to proceed.	canContinue false

11. Failure cases

- bobPatchResult.json missing or status is patch-scope-failed, invalidated, or failed.
- Any immutable artifact hash mismatch — produces invalidated and stops before Playwright runs.
- Playwright fails to start or JSON output is unparseable — produces infra-failed.
- Bob introduced new failures (newFailures is non-empty) — reported honestly in evidenceReport.md, status remains passed or failed based on overall test count, but new failures are highlighted.

12. Definition of done

- Phase 8 reads bobPatchResult.json and refuses to proceed if status does not allow continuation.
- Immutable artifact hashes are re-verified against Phase 5 baseline values.
- Playwright reruns the same test file with the same command as Phase 6.
- Final results are mapped to locked-spec check IDs.
- Resolved and new-failure sets are computed from the baseline comparison.
- finalVerificationResult.json is written with status passed or failed (not infra-failed or invalidated).

- evidenceReport.md is written with all eight sections, manual review items, and claim boundary.
- The allowed conclusion is Passed SpecFirst automated contract. Manual review required. The forbidden claims are never present in the report.
- Phase 8 does not patch code, call Bob, modify any artifact, or suppress failures.

13. Final implementation note

Phase 8 is the moment the pipeline proves its value. It shows, in a single evidence report, what the component looked like before Bob, what Bob changed, what the tests prove now, and what still requires human judgment. The claim boundary is not a disclaimer — it is the honest boundary between what a deterministic accessibility contract can certify and what only human assistive technology testing can confirm.

End of Phase 8 documentation.